Linear Algebra and Its Applications

Section 4.2: Null Spaces, Column Spaces, and Linear Transformations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.2 Null spaces, column spaces, and linear transformations

For an $m \times n$ matrix A, the null space $\operatorname{Nul} A$ contains the *inputs* $x \in \mathbb{R}^n$ satisfying $Ax = 0$. The column space $\operatorname{Col} A$ contains the *outputs* $Ax \in \mathbb{R}^m$. Do not confuse these two roles. To test $w \in \operatorname{Nul} A$, multiply Aw and compare with zero. To test $w \in \operatorname{Col} A$, solve $Ax = w$ and ask whether a solution exists. A homogeneous linear equation defines a subspace; a nonhomogeneous equation usually describes a translated subspace and does not even contain zero. For a linear map $T : V \to W$, the corresponding terms are kernel $\{v : T(v) = 0\}$ and range $\{T(v) : v \in V\}$.

Practice problem 4.2.P1: Two descriptions of one plane

For $W = \{(a, b, c) : a - 3b - c = 0\}$, first observe that $W = \operatorname{Nul}[\,1 \;\; -3 \;\; -1\,]$. The null space theorem therefore makes it a subspace of $\mathbb{R}^3$. Alternatively solve $a = 3b + c$. Then

$$(a, b, c) = b(3, 1, 0) + c(1, 0, 1).$$

As b, c vary freely, this proves $W = \operatorname{Span}\{(3, 1, 0), (1, 0, 1)\}$, a subspace by the span theorem. The first description is implicit (an equation); the second is explicit (generators and arbitrary coefficients).

Practice problem 4.2.P2: Add two attainable outputs

Suppose $Ax = v$ and $Ay = w$ have solutions x, y. By distributivity, $A(x + y) = Ax + Ay = v + w$, so the equation with right-hand side $v + w$ is also consistent. Equivalently $v, w \in \operatorname{Col} A$, and the column space is closed under addition. We need not find either of the original solutions.

Exercise 4.2.1: Test a proposed null-space vector directly

What we must show. Decide whether $w = (1, 3, -4)$ is in $\operatorname{Nul} A$ for the displayed matrix. Multiply using the original rows, retaining the signs:

$$\begin{pmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 - 15 + 12 \\ 6 - 6 + 0 \\ -8 + 12 - 4 \end{pmatrix} = 0.$$

Every coordinate is zero, so $w \in \operatorname{Nul} A$. A vector belongs to the null space precisely when this multiplication gives zero, not merely when its coordinates resemble a row or column of A.

Exercise 4.2.2: Another direct multiplication

With $w = (1, -1, 1)$, compute

$$\begin{pmatrix} 2 & 6 & 4 \\ -3 & 2 & 5 \\ -5 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 - 6 + 4 \\ -3 - 2 + 5 \\ -5 + 4 + 1 \end{pmatrix} = 0.$$

Thus $w \in \operatorname{Nul} A$. All three equations must hold; checking only the first row would not have been enough.

Exercise 4.2.3: Describe a null space using two free variables

The homogeneous system for $A = \left(\begin{smallmatrix} 1 & 2 & 4 & 0 \\ 0 & 1 & 3 & -2 \end{smallmatrix}\right)$ is

$$x_1 + 2x_2 + 4x_3 = 0, \qquad x_2 + 3x_3 - 2x_4 = 0.$$

The pivot variables are x_1, x_2; choose $x_3 = s, x_4 = t$ freely. The second equation gives $x_2 = -3s + 2t$, and substitution in the first gives $x_1 = -2(-3s + 2t) - 4s = 2s - 4t$. Therefore

$$x = s \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -4 \\ 2 \\ 0 \\ 1 \end{pmatrix}.$$

Both parameters may be any real number, so the two displayed vectors span $\operatorname{Nul} A$; this formula describes every solution and only solutions.

Exercise 4.2.4: A forced zero coordinate and two free coordinates

For $A = \left(\begin{smallmatrix} 1 & -3 & 2 & 0 \\ 0 & 0 & 3 & 0 \end{smallmatrix}\right)$, the second equation is $3x_3 = 0$, so $x_3 = 0$. The first becomes $x_1 = 3x_2$. Letting $x_2 = s$ and $x_4 = t$ gives

$$\operatorname{Nul} A = \left\{ s \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

In particular the fourth column being zero leaves the fourth input free; it does not require the fourth input to be zero.

Exercise 4.2.5: Back substitution in a five-column matrix

The nonzero rows give

$$x_1 - 4x_2 + 2x_4 = 0, \qquad x_3 - 5x_4 = 0, \qquad 2x_5 = 0.$$

The last equation gives $x_5 = 0$. Set the free variables $x_2 = s, x_4 = t$. Then $x_3 = 5t$ and $x_1 = 4s - 2t$. Thus

$$\operatorname{Nul} A = \operatorname{Span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Each generating vector is obtained by giving one free variable value 1 and the other value 0; their arbitrary combinations preserve both free variables.

Exercise 4.2.6: Three free variables

The matrix gives two effective equations:

$$x_1 + 3x_2 - 4x_3 - 3x_4 + x_5 = 0, \qquad x_2 - 3x_3 + x_4 = 0.$$

The zero third row adds no condition. Set $x_3 = r, x_4 = s, x_5 = t$. Then $x_2 = 3r - s$ and $x_1 = -3(3r - s) + 4r + 3s - t = -5r + 6s - t$. Consequently

$$\operatorname{Nul} A = \operatorname{Span}\left\{\begin{pmatrix}-5\\3\\1\\0\\0\end{pmatrix}, \begin{pmatrix}6\\-1\\0\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\0\\1\end{pmatrix}\right\}.$$

Exercise 4.2.7: An affine plane misses zero

The equation is $a+b+c=2$. At $(a,b,c)=(0,0,0)$ it would read $0=2$, which is false. Hence the zero vector is not in W; W is not a vector space under the usual operations. The fact that the equation is linear in the variables does not make its *nonhomogeneous* solution set a subspace.

Exercise 4.2.8: Move all variable terms to one side

The relation $3r - 2 = 3s + t$ becomes $3r - 3s - t = 2$. Zero does not satisfy this relation, because its left side is zero. Therefore the given set is not a vector space.

Exercise 4.2.9: Two homogeneous constraints define a null space

Rewrite $p - 3q = 4s$ and $2p = s + 5r$ as

$$p - 3q + 0r - 4s = 0, \qquad 2p + 0q - 5r - s = 0.$$

Thus W is the null space of $\left(\begin{smallmatrix}1 & -3 & 0 & -4\\ 2 & 0 & -5 & -1\end{smallmatrix}\right)$, and the null space theorem makes it a subspace of $\mathbb{R}^4$. Both right-hand sides after rearrangement are zero; this is the essential property.

Exercise 4.2.10: Another homogeneous description

Move all terms to the left: $3a + b - c + 0d = 0$ and $a + b + 2c - 2d = 0$. Then

$$W = \operatorname{Nul}\begin{pmatrix}3 & 1 & -1 & 0\\ 1 & 1 & 2 & -2\end{pmatrix}.$$

The null space of a matrix is a subspace of its input space, here $\mathbb{R}^4$. This proves the requested vector-space property without finding parameters.

Exercise 4.2.11: A hidden constant obstructs the zero vector

For $(s - 2t, 3 + 3s, 3s + t, 2s)$ to be zero, its fourth coordinate forces $s = 0$. But then its second coordinate is 3, not zero. No choice of t can fix that contradiction. Thus zero is absent, and W is not a vector space. Merely setting all parameters zero would give only one candidate; the argument above checks that *no* parameters produce zero.

Exercise 4.2.12: Another parametrization that never produces zero

If $(3p-5q, 4q, p, q+1) = 0$, the third coordinate forces $p = 0$ and the second forces $q = 0$. The fourth would then be 1. This contradiction shows that zero cannot occur, so W is not a vector space.

Exercise 4.2.13: A span in disguise

For arbitrary real c, d,

$$(c-6d, d, c) = c(1,0,1) + d(-6,1,0).$$

Both directions of this parametrization hold, so $W = \text{Span}\{(1,0,1), (-6,1,0)\}$. The span theorem makes it a subspace of $\mathbb{R}^3$.

Exercise 4.2.14: A two-generator subspace

Collecting the coefficients of s, t gives

$$(-s+3t, s-2t, 5s-t) = s(-1,1,5) + t(3,-2,-1).$$

Since s, t are unrestricted real numbers, W is exactly the span of these two vectors, hence is a subspace of $\mathbb{R}^3$.

Exercise 4.2.15: Make the coefficient vectors into columns

What we must show. Find a matrix whose column space is the given parametrized set. Separate the coefficients of r, s, t:

$$\begin{pmatrix} 2s+t \\ r-s+2t \\ 3r+s \\ 2r-s-t \end{pmatrix} = r\begin{pmatrix} 0 \\ 1 \\ 3 \\ 2 \end{pmatrix} + s\begin{pmatrix} 2 \\ -1 \\ 1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix}.$$

Therefore one suitable matrix is

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \\ 2 & -1 & -1 \end{pmatrix}.$$

By definition its column space contains exactly the linear combinations just listed. Other matrices can have the same column space; uniqueness is not requested.

Exercise 4.2.16: Construct a matrix from three parameters

Here the required equality is

$$\begin{pmatrix} b-c \\ 2b+3d \\ b+3c-3d \\ c+d \end{pmatrix} = b\begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} -1 \\ 0 \\ 3 \\ 1 \end{pmatrix} + d\begin{pmatrix} 0 \\ 3 \\ -3 \\ 1 \end{pmatrix}.$$

Thus take $A = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 3 \\ 1 & 3 & -3 \\ 0 & 1 & 1 \end{pmatrix}$. For $x = (b, c, d)$, the product Ax gives the original form, and every such form is obtained from some x. This proves equality with $\operatorname{Col} A$.

Exercise 4.2.17: Count inputs and outputs

The displayed matrix has four rows and two columns, so its mapping is $A : \mathbb{R}^2 \to \mathbb{R}^4$.

Conclusion. (a) $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^2$, so $k = 2$. (b) $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, so $k = 4$.

These are ambient-space dimensions, not necessarily the dimensions of the two subspaces.

Exercise 4.2.18: A rectangular matrix with three inputs

The matrix has four rows and three columns. A vector x in Ax must therefore have three coordinates, whereas the output has four.

Conclusion. (a) $k = 3$ for the ambient space of $\operatorname{Nul} A$. (b) $k = 4$ for the ambient space of $\operatorname{Col} A$.

Exercise 4.2.19: Five inputs and two outputs

The matrix has size 2×5, so Ax is defined for $x \in \mathbb{R}^5$ and belongs to $\mathbb{R}^2$.

Conclusion. (a) $\operatorname{Nul} A \subseteq \mathbb{R}^5$, so $k = 5$. (b) $\operatorname{Col} A \subseteq \mathbb{R}^2$, so $k = 2$.

Exercise 4.2.20: A row matrix

Here $A = [\,1 \;\; -3 \;\; 2 \;\; 0 \;\; -5\,]$ has one row and five columns. It maps a five-coordinate input to one real number.

Conclusion. (a) $k = 5$ for $\operatorname{Nul} A$. (b) $k = 1$ for $\operatorname{Col} A$.

Exercise 4.2.21: Exhibit one vector in each space

Using the matrix in Exercise 4.2.17, its second column equals $-2/3$ times its first. Therefore $2a_1 + 3a_2 = 0$, giving the nonzero null vector $(2, 3)$. A direct check is

$$\begin{pmatrix} 6 & -4 \\ -3 & 2 \\ -9 & 6 \\ 9 & -6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 - 12 \\ -6 + 6 \\ -18 + 18 \\ 18 - 18 \end{pmatrix} = 0.$$

The first column $(6, -3, -9, 9)$ is a nonzero member of $\operatorname{Col} A$. It equals $A(1, 0)$, so its membership follows directly from the definition.

Exercise 4.2.22: A null relation among three columns

For the matrix in Exercise 4.2.18, choose $x = (-1, -1, 1)$. Multiplication verifies

$$Ax = \begin{pmatrix} -5+2+3 \\ 1+0-1 \\ 0+2-2 \\ 5-7+2 \end{pmatrix} = 0.$$

Thus this is a nonzero vector of $\operatorname{Nul} A$. The first column $(5, -1, 0, -5)$ is a nonzero vector in $\operatorname{Col} A$, because it is the output $A(1, 0, 0)$.

Exercise 4.2.23: A vector can belong to both spaces

For $A = \left(\begin{smallmatrix} -2 & 4 \\ -1 & 2 \end{smallmatrix}\right)$ and $w = (2, 1)$, the negative of the first column is w. Hence $A(-1, 0) = w$, proving $w \in \operatorname{Col} A$. For null-space membership test the different multiplication:

$$Aw = (-4+4, -2+2) = (0, 0).$$

Thus $w \in \operatorname{Nul} A$ as well. These two statements are compatible; a square matrix's input and output spaces have the same ambient space, though they serve different roles.

Exercise 4.2.24: Two independent membership checks in four dimensions

The matrix and vector are

$$A = \begin{pmatrix} 10 & -8 & -2 & -2 \\ 0 & 2 & 2 & -2 \\ 1 & -1 & 6 & 0 \\ 1 & 1 & 0 & -2 \end{pmatrix}, \qquad w = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 2 \end{pmatrix}.$$

Column-space membership.

The first two columns add to w, so $A(1, 1, 0, 0) = w$. This is a concrete solution of $Ax = w$, proving $w \in \operatorname{Col} A$.

Null-space membership.

Use w itself as the input:

$$Aw = (20 - 16 - 4,\ 4 - 4,\ 2 - 2,\ 2 + 2 - 4) = (0, 0, 0, 0).$$

Conclusion. w belongs to both $\operatorname{Col} A$ and $\operatorname{Nul} A$.

Exercise 4.2.25: Six statements about null spaces and column spaces

(a) **True.** $\operatorname{Nul} A = \{x : Ax = 0\}$ is the definition.

(b) **False.** An $m \times n$ matrix has n input coordinates, so its null space is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(c) **True.** The possible outputs of $x \mapsto Ax$ are precisely all combinations of the columns, hence form $\operatorname{Col} A$.

(d) **False.** One consistent equation only says that this particular b is in the column space. For instance $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $b = (1, 0)$ give a consistent equation but $\operatorname{Col} A$ is just the horizontal axis.

(e) **True.** The kernel contains zero and is closed under addition and scalar multiplication because the map is linear.

(f) **True.** $\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\}$; this is the output description of the span of the columns.

Exercise 4.2.26: Six related statements about kernels and ranges

(a) **True.** A null space is a subspace of the input space, hence a vector space.

(b) **True.** Columns of an $m \times n$ matrix have m coordinates; their combinations belong to $\mathbb{R}^m$.

(c) **False.** Solutions of $Ax = b$ are inputs x, not outputs. The column space instead contains those right-hand sides b for which a solution exists.

(d) **True.** The kernel of $x \mapsto Ax$ is, by definition, the set of inputs taken to zero, exactly $\operatorname{Nul} A$.

(e) **True.** The range is a subspace of the target vector space: add two outputs or multiply an output by a scalar and linearity produces another output.

(f) **True.** A homogeneous linear differential equation is $L(f) = 0$ for a linear differential operator L. Its solutions are the kernel of L on the vector space of functions with the required derivatives.

Exercise 4.2.27: Scale a known homogeneous solution

The given solution vector is $u = (3, 2, -1)$. The proposed vector is $10u$. The solution set of the homogeneous system is a null space, which is a vector space and is therefore closed under scalar multiplication. Consequently $10u = (30, 20, -10)$ is another solution. This uses no substitution or row operations, as the exercise requests.

Exercise 4.2.28: Scale an attainable right-hand side

Both systems have the same coefficient matrix A. Their right-hand sides are $b = (0, 1, 9)$ and $5b = (0, 5, 45)$, respectively. Since the first is consistent, there is an input u with $Au = b$. Linearity gives $A(5u) = 5Au = 5b$, so the second system is consistent too. Equivalently $b \in \operatorname{Col} A$ and column spaces are closed under scalar multiplication.

Exercise 4.2.29: Prove the column-space subspace theorem

A general member of $\operatorname{Col} A$ has form Ax, where $x \in \mathbb{R}^n$.

(a) Zero is an output.

Take $x = 0$. Then $Ax = A0 = 0$, so the zero vector of $\mathbb{R}^m$ lies in $\operatorname{Col} A$.

(b) Two outputs can be added.

For inputs $x, w \in \mathbb{R}^n$, distributivity gives $Ax + Aw = A(x + w)$. Since $x + w$ is another allowed input, this is an element of $\operatorname{Col} A$.

(c) An output can be multiplied by any scalar.

For real c, the identity $c(Ax) = A(cx)$ expresses it as an output as well.

Conclusion. The three conditions prove that $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Exercise 4.2.30: The range of a linear transformation is a subspace

Let $R = \{T(x) : x \in V\} \subseteq W$. First $T(0) = 0$, so $0 \in R$. For arbitrary $y = T(x)$ and $z = T(v)$ in R, linearity gives

$$y + z = T(x) + T(v) = T(x + v) \in R.$$

For any real scalar c, $cy = cT(x) = T(cx) \in R$. The inputs $x + v, cx$ are in V because V is a vector space. Thus R passes all subspace tests inside W. One need not assume that T is onto W.

Exercise 4.2.31: Evaluate a quadratic at two inputs

The map is $T(p) = (p(0), p(1))$ from P_2 into $\mathbb{R}^2$.

(a) Check the two linearity identities at both coordinates.

For polynomials p, q and scalar c,

$$T(p + q) = (p(0) + q(0), p(1) + q(1)) = T(p) + T(q), \qquad T(cp) = cT(p).$$

Thus T is linear.

(b) Solve the kernel equations and then describe the range.

Write $p(t) = a + bt + ct^2$. The equations $p(0) = p(1) = 0$ give $a = 0$ and $b + c = 0$, hence $p(t) = c(t^2 - t)$. The kernel is therefore $\operatorname{Span}\{t^2 - t\}$. For any desired output (r, s), the polynomial $p(t) = r + (s - r)t$ satisfies $p(0) = r$ and $p(1) = s$. Therefore the range is all of $\mathbb{R}^2$.

Conclusion. One kernel generator is $t^2 - t$, and T is onto $\mathbb{R}^2$.

Exercise 4.2.32: Repeat the same evaluation twice

Here $T(p) = (p(0), p(0))$. With $p(t) = a + bt + ct^2$, the output is (a, a). It is zero exactly when $a = 0$, leaving b, c arbitrary. Thus

$$\ker T = \operatorname{Span}\{t, t^2\}, \qquad \operatorname{range} T = \{(a, a) : a \in \mathbb{R}\} = \operatorname{Span}\{(1, 1)\}.$$

The output has two coordinates, but those coordinates must be equal; therefore the map is not onto the whole target $\mathbb{R}^2$.

Exercise 4.2.33: Add a matrix to its transpose

Define $T(A) = A + A^T$ on $M_{2\times 2}$.

(a) Verify linearity using transpose rules.

For matrices A, C and real scalar r,

$$T(A+C) = A + C + (A+C)^T = (A + A^T) + (C + C^T), \qquad T(rA) = r(A + A^T).$$

These are exactly the two linearity identities.

(b) Produce a preimage of any symmetric matrix.

If $B^T = B$, choose $A = B/2$. Then $T(A) = B/2 + B^T/2 = B$.

(c) Establish equality of the range with the symmetric matrices.

Every output satisfies $(A + A^T)^T = A^T + A = A + A^T$, so every output is symmetric. Conversely part (b) realizes every symmetric matrix as an output. Both inclusions prove the claimed equality of sets.

(d) Solve the zero-output condition entry by entry.

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$T(A) = \begin{pmatrix} 2a & b+c \\ b+c & 2d \end{pmatrix}.$$

This is zero exactly when $a = d = 0$ and $c = -b$. Therefore

$$\ker T = \left\{ b \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

Exercise 4.2.34: The antiderivative with a fixed initial value

For continuous f on $[0, 1]$, the specified antiderivative is uniquely

$$T(f)(x) = \int_0^x f(t)\, dt.$$

Existence follows from the Fundamental Theorem of Calculus. For uniqueness, two such antiderivatives differ by a constant, and their value zero at the left endpoint forces that constant to be zero. Linearity of the integral gives $T(f+g)(x) = T(f)(x) + T(g)(x)$ and $T(cf)(x) = cT(f)(x)$ for every x. Thus these are equalities of functions, and T is a linear transformation. If $T(f)$ is the zero function, differentiating it gives $f(x) = 0$ on $(0, 1)$, and continuity gives zero at the endpoints too. Conversely the zero function has zero antiderivative with the prescribed initial value.

Conclusion. The kernel consists only of the zero function.

Exercise 4.2.35: The image of a subspace

Let U be a subspace of V. Since $0 \in U$ and $T(0) = 0$, zero is in $T(U)$. For $y = T(u)$ and $z = T(v)$ with $u, v \in U$, linearity gives $y + z = T(u+v) \in T(U)$ because $u + v \in U$. Likewise $cy = T(cu) \in T(U)$ since $cu \in U$ for every real c. These three tests prove that $T(U)$ is a subspace of W. The conclusion concerns the image of the whole subspace, not just a list of selected outputs.

Exercise 4.2.36: The inverse image of a subspace

Let $U = \{x \in V : T(x) \in Z\}$, where Z is a subspace of W. Since $T(0) = 0 \in Z$, zero belongs to U. If $x, y \in U$, then $T(x), T(y) \in Z$, and $T(x + y) = T(x) + T(y) \in Z$ by additive closure of Z. Thus $x + y \in U$. For real c, $T(cx) = cT(x) \in Z$, so $cx \in U$. Therefore U is a subspace of V. This argument uses the preimage as a set; it does not require T to have an inverse function.

Exercise 4.2.37: An output that is not a zero-output input

The displayed data are

$$A = \begin{pmatrix} 7 & 6 & -4 & 1 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 1 \end{pmatrix}, \qquad w = (1, 1, -1, -3).$$

To test null-space membership, multiply:

$$Aw = (7 + 6 + 4 - 3,\ -5 - 1 + 6,\ 9 - 11 - 7 + 9,\ 19 - 9 - 7 - 3) = (14, 0, 0, 0) \neq 0.$$

Therefore $w \notin \operatorname{Nul} A$. For the column space, the second column plus the third minus the fourth is

$$(6, -1, -11, -9) + (-4, 0, 7, 7) - (1, -2, -3, 1) = (1, 1, -1, -3) = w.$$

Thus $A(0, 1, 1, -1) = w$, which proves $w \in \operatorname{Col} A$.

Conclusion. w is in the column space only.

Exercise 4.2.38: Another vector lying in both spaces

Here

$$A = \begin{pmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{pmatrix}, \qquad w = (1, 2, 1, 0).$$

First $Aw = (-8 + 10 - 2,\ -5 + 4 + 1,\ 10 - 16 + 6,\ 3 - 4 + 1) = 0$, so $w \in \operatorname{Nul} A$. For the distinct column-space test use $x = (-2, -3, 0, 1)$:

$$Ax = (16 - 15,\ 10 - 6 - 2,\ -20 + 24 - 3,\ -6 + 6) = (1, 2, 1, 0) = w.$$

Conclusion. w belongs to both spaces, verified by two different products.

Exercise 4.2.39: Recover column relations and the entire null space

The given matrix is

$$A = \begin{pmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{pmatrix}, \qquad B = [\, a_1\ a_2\ a_4 \,].$$

(a) Exhibit the two column combinations.

Coordinate-wise verification gives

$$a_3 = \tfrac{1}{3}a_1 + \tfrac{1}{3}a_2, \qquad a_5 = \tfrac{10}{3}a_1 - \tfrac{26}{3}a_2 - 4a_4.$$

Both are combinations of columns of B, so $a_3, a_5 \in \operatorname{Col} B$.

(b) Solve all homogeneous equations.

Exact row reduction gives

$$A \sim \begin{pmatrix} 1 & 0 & 1/3 & 0 & 10/3 \\ 0 & 1 & 1/3 & 0 & -26/3 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Take the free variables $x_3 = s, x_5 = t$. Then $x_1 = -s/3 - 10t/3$, $x_2 = -s/3 + 26t/3$, and $x_4 = 4t$. Hence

$$\operatorname{Nul} A = \operatorname{Span}\left\{ \begin{pmatrix} -1/3 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -10/3 \\ 26/3 \\ 0 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

These formulas also check the two column relations in part (a): moving all terms to one side gives the corresponding zero combinations.

(c) Explain both failures of invertibility separately.

The first null-space generator is nonzero, so T is not one-to-one: it maps that generator and zero to the same output. There are only three pivots among four rows, so the columns do not span $\mathbb{R}^4$; hence T is not onto $\mathbb{R}^4$. Its having five inputs does not by itself make it onto.

Exercise 4.2.40: Find the line where two planes meet

What we must show. Construct a nonzero vector in both spans and show that its span is the entire intersection. Let $w = c_1v_1 + c_2v_2 = c_3v_3 + c_4v_4$. Move the last two terms left and solve the homogeneous system with columns $v_1, v_2, -v_3, -v_4$:

$$\begin{pmatrix} 5 & 1 & -2 & 0 \\ 3 & 3 & 1 & 12 \\ 8 & 4 & -5 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -10/3 \\ 0 & 1 & 0 & 26/3 \\ 0 & 0 & 1 & -4 \end{pmatrix}.$$

There is one free variable, $c_4 = t$, and $c_1 = 10t/3$, $c_2 = -26t/3$, $c_3 = 4t$. Taking $t = 1$ gives

$$w = \tfrac{10}{3}(5, 3, 8) - \tfrac{26}{3}(1, 3, 4) = (8, -16, -8).$$

Independently the other plane gives $4(2, -1, 5) + (0, -12, -28) = (8, -16, -8)$, confirming both memberships. Every possible set of coefficients in the homogeneous system is a multiple of this one, so every common vector is a multiple of w.

Conclusion. $H \cap K = \operatorname{Span}\{(8, -16, -8)\} = \operatorname{Span}\{(1, -2, -1)\}$.